Program:2(**MySQL Database Creation, Table Creation, Query – Student Database**)

```
CREATE DATABASE college_db;
USE college_db;
CREATE TABLE student (     student_id INT PRIMARY KEY,
   student_name VARCHAR(50),
  department VARCHAR(30),
  age INT,     marks INT      );
INSERT INTO student VALUES     (101, 'Anitha', 'MCA', 22, 85),
 (102, 'Banu', 'MCA', 23, 90),
(103, 'Chitra', 'MCA', 22, 78),
 (104, 'Divya', 'MCA', 23, 88);
SELECT * FROM student;
SELECT student_id, student_name, marks FROM student;
SELECT * FROM student WHERE marks > 80;
UPDATE student SET marks = 92 WHERE student_id = 102;
DELETE FROM student WHERE student_id = 103;
ALTER TABLE student ADD email VARCHAR(60);
ALTER TABLE student DROP COLUMN email;
TRUNCATE TABLE student;
SELECT * FROM student;
```

Programm:4 (**Spatial data storage and retrieval in MySQL Command Line Client**)

```
CREATE DATABASE sp1;
USE sp1;

create table spat(scode int,des varchar(20),geo geometry);
set @g='linestring(0 0,2 2,4 4)';
insert into spat(scode,des,geo)values(2001,"data",ST_GeomFromText(@g));
select * from spat;
alter table spat add ploca point;
alter table spat drop ploca;
desc spat;
alter table spat modify des varchar(10);
desc spat;
select scode,des,ST_AsText(geo) from spat;
```

program:5(**Temporal data storage and retrieval in MySQL Command Line Client**)

```
select now();
select curdate();
select curtime();
select current_date();
select current_time();
select current_timestamp();
select sysdate();
select sec_to_time(200);
select datediff('2023-09-06','2023-04-30');
select adddate('2023-04-30',129);
select subdate('2023-09-06',129);
select timediff('12:30:30','11:30:30');
select addtime('1:30:30','11:30:30');
select subtime('12:30:30','11:30:30');
```

program:6 (**Object Storage and Retrieval in MYSQL**)

```
CREATE DATABASE college_db1;

USE college_db1;

CREATE TABLE student(s_id INT AUTO_INCREMENT PRIMARY KEY,s_name varchar(75),dept varchar(25),cont_det json);

INSERT INTO student (s_name, dept, cont_det)values('Alisha','MCA','{ "place": "Vellore","Mail_id":"Alisha@gmail.com","phone_no":1234567890}');

INSERT INTO student (s_name, dept, cont_det)values('Aaliya','MCA','{ "place": "Ambur","Mail_id":"Aaliya@gmail.com","phone_no":9087654321}');

INSERT INTO student (s_name, dept, cont_det)values('Wassim','MCA','{ "place": "Gudiyattam","Mail_id":"Wassim@gmail.com","phone_no":8907654321 }');

INSERT INTO student (s_name, dept, cont_det)values('Alaiya','MCA','{ "place":"Gudiyattam","Mail_id":"Alaiya@gmail.com","phone_no":7890654321}' );

select * from student;

SELECT s_id,s_name, cont_det->'$.Mail_id' AS contact_deatils FROM student;
```